Inheritance

Module-1



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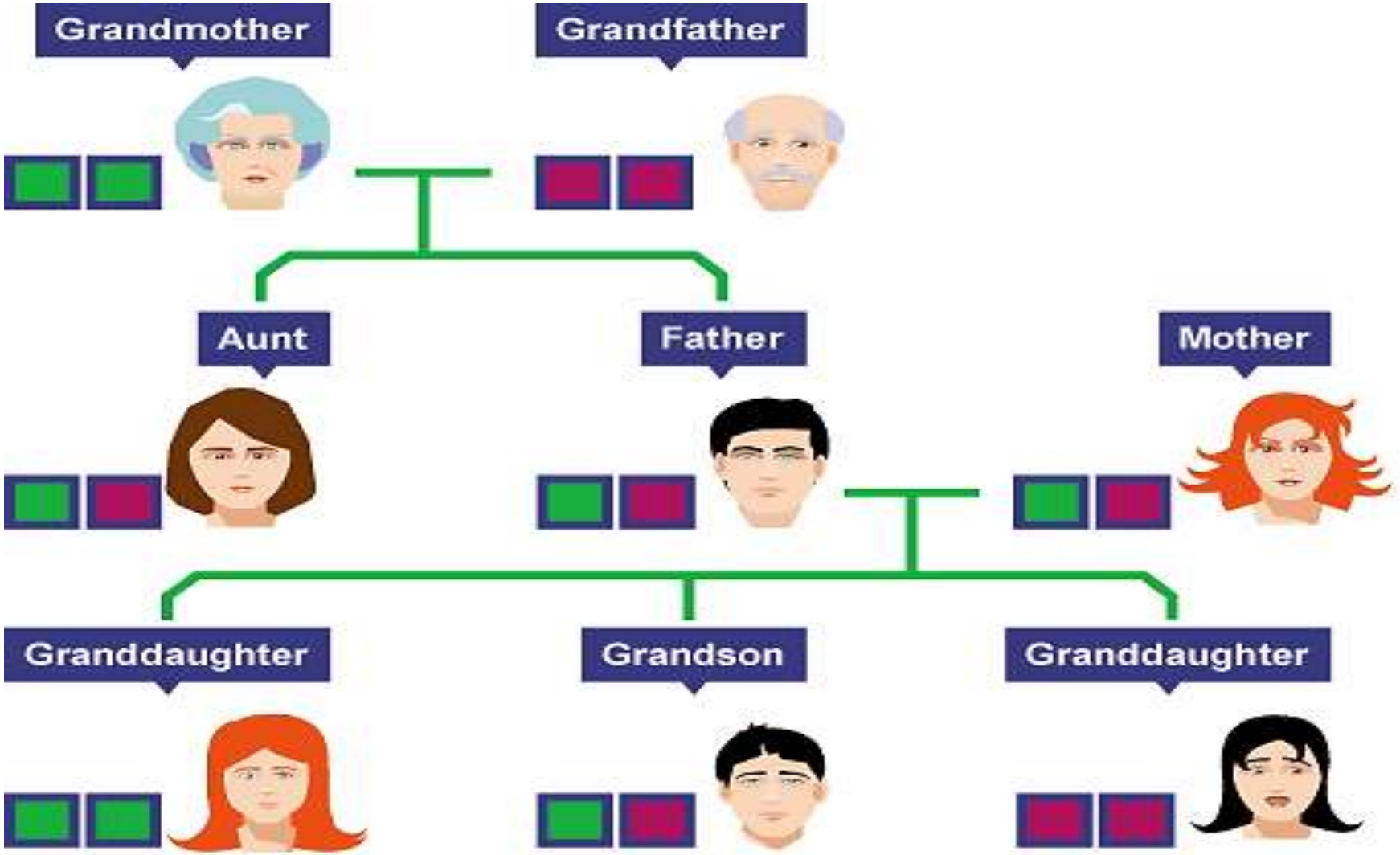
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Outline

- ✓ Inheritance Concept
- ✓ Types of Inheritance

Inheritance

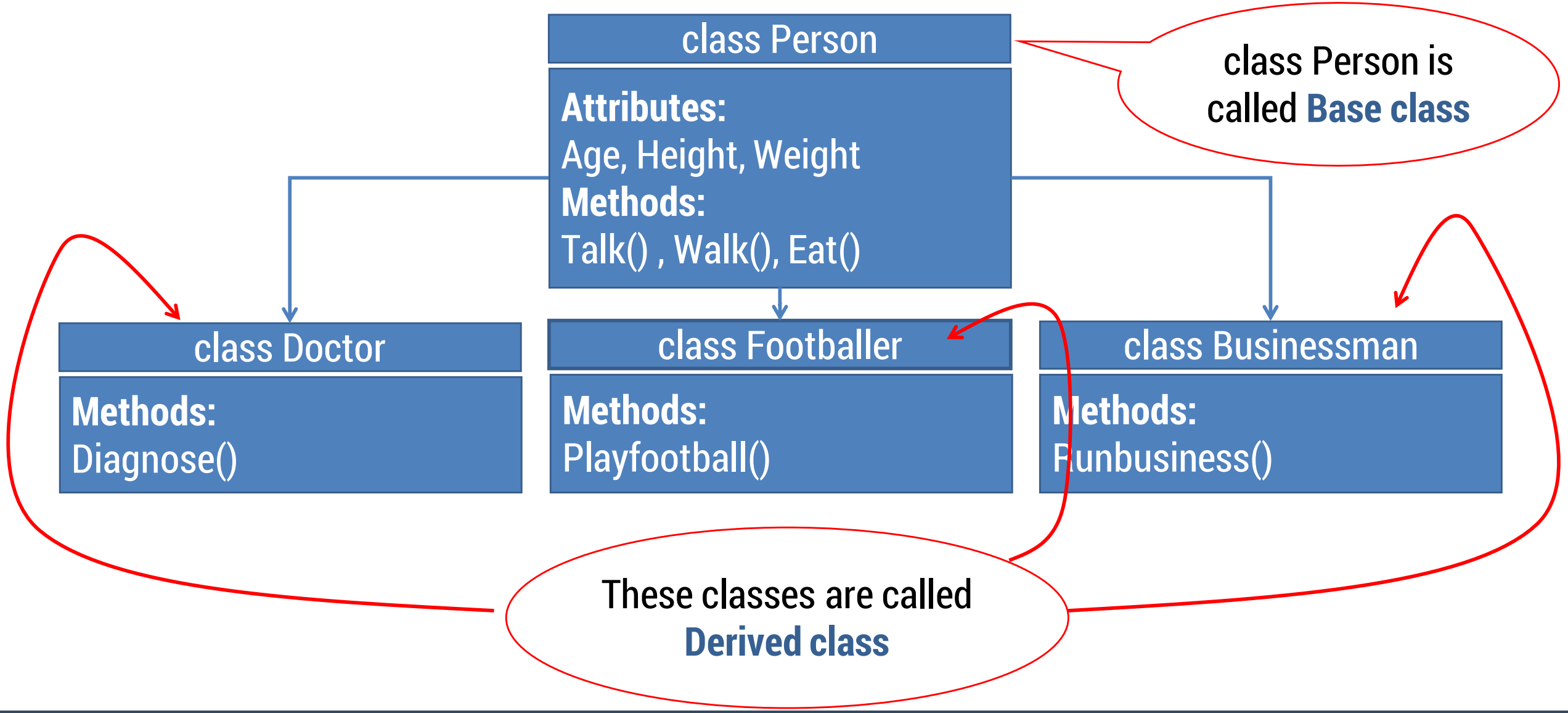


Concept of Inheritance

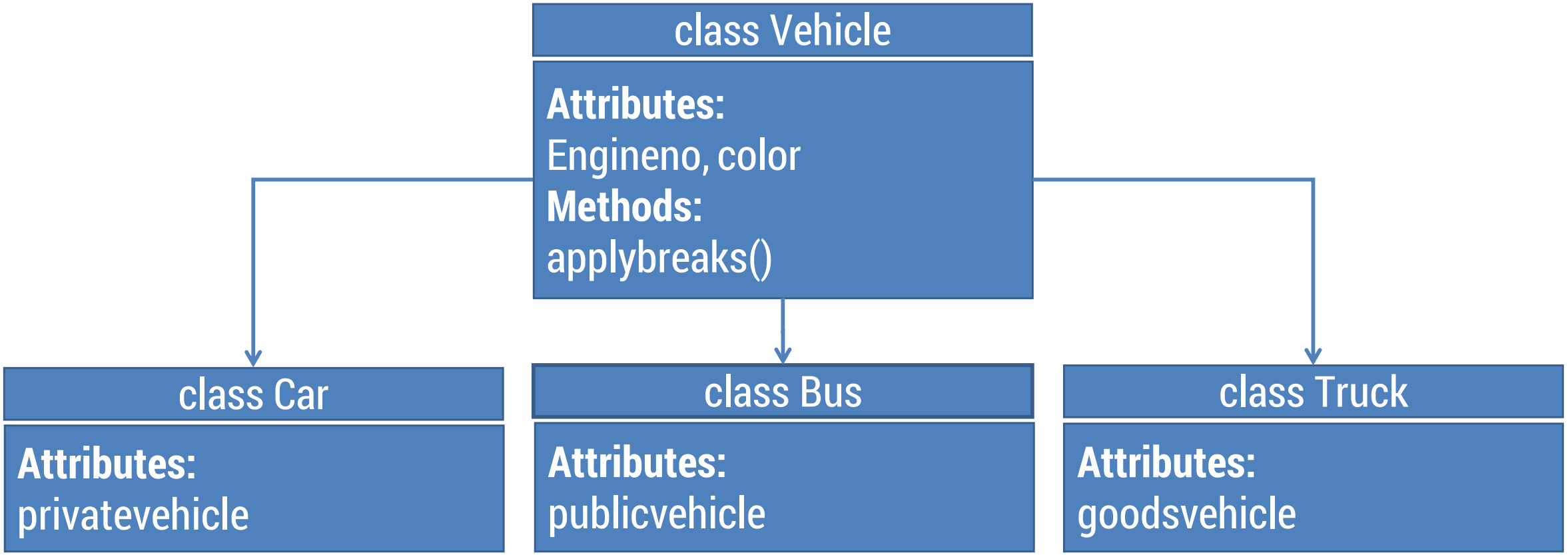
class Doctor	class Footballer	class Businessman
Attributes: Age, Height, Weight **Methods:** Talk() Walk() Eat() Diagnose()	**Attributes:** Age, Height, Weight **Methods:** Talk() Walk() Eat() Playfootball()	**Attributes:** Age, Height, Weight **Methods:** Talk() Walk() Eat() Runbusiness()

- All of the classes have common attributes (Age, Height, Weight) and methods (Walk, Talk, Eat).
- However, they have some special skills like Diagnose, Playfootball and Runbusiness.
- In each of the classes, you would be copying the same code for Walk, Talk and Eat for each character.

Concept of Inheritance(Cont…)



Concept of Inheritance(Cont…)



Inheritance

- **Inheritance** is the process, by which class can acquire(reuse) the properties and methods of another class.
- The mechanism of deriving a new class from an old class is called **inheritance**.
- The new class is called **derived class** and old class is called **base class**.
- The derived class may have all the features of the base class and the programmer can add new features to the derived class.

Inheritance

Advantages:

- Application development and execution time is less with low memory.
- **Redundancy** (repetition) of the code is reduced or minimized so that we get consistence results and less storage cost.
- Another reason is the idea of **reusability.** One can derive a new class (sub-class) from an existing class and add new features to it without modifying its parent class. There is no need to rewrite the parent class in order to inherit it.
- One reason is the **transitive nature.** If class **A** inherits properties from another class **B**, then all subclasses of **A** will automatically inherit properties from **B**. This property is called the transitive nature of inheritance.

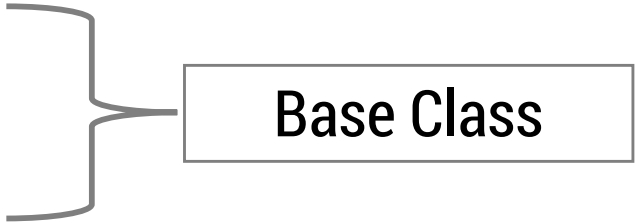
Syntax to Inherit class

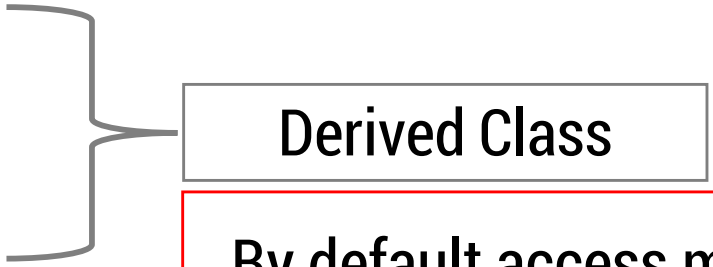
Syntax:

```
class derived-class-name extends base-class-name
{
      // body of class
}
```

Example:

```
class person{
   //body of class
}
class doctor extends person
{
  //body of class
}
```



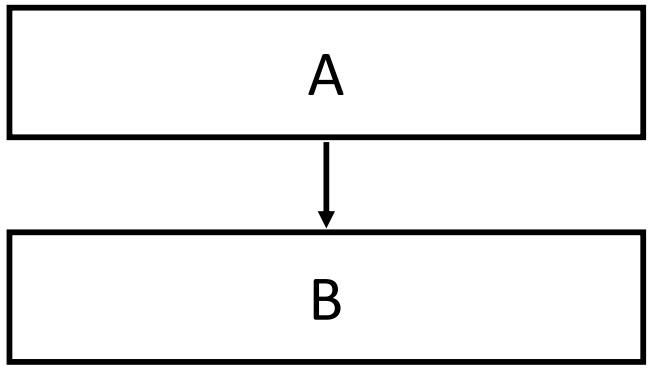


By default access mode is ‘default’

Types of Inheritance

1. Single Inheritance
2. Multilevel Inheritance
3. Multiple Inheritance
4. Hierarchical Inheritance
5. Hybrid Inheritance

1. Single Inheritance



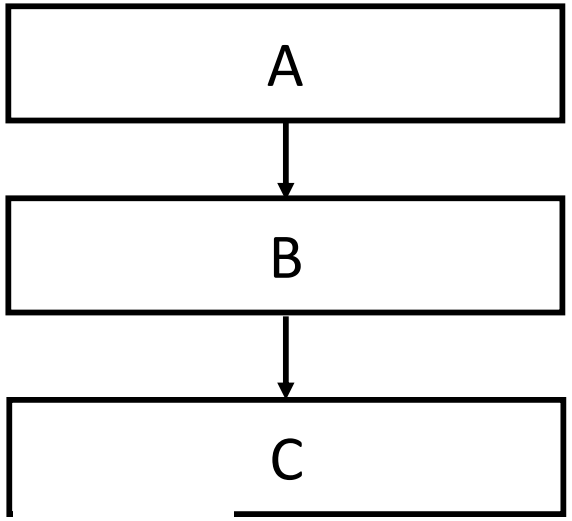
- If a class is derived from a single class then it is called **single inheritance**.
- Class **B** is derived from class **A**

```
class Faculty
{
float salary=30000;
}
class Science extends Faculty
{
float bonous=2000;
public static void main(String args[])
{
Science obj=new Science();
System.out.println("Salary is:"+obj.salary);
System.out.println("Bonous is:"+obj.bonous);
}
}
```

Output

```
Salary is: 30000.0
Bonous is: 2000.0
```

2. Multilevel Inheritance



- Any class is derived from a class which is derived from another class then it is called **multilevel inheritance**.
- Here, class **C** is derived from class **B** and class **B** is derived from class **A**, so it is called **multilevel inheritance**.

Example:

```
class Person
{
   //content of class person
}
class Student extends Person
{
  //content of Student class
}
```

```
class ITStudent extends Student
{
   //content of ITStudent  class
}
```

2. Multilevel Inheritance

```
class Faculty
{
float total_sal=0, salary=30000;
}

class HRA extends Faculty
{
float hra=3000;
}

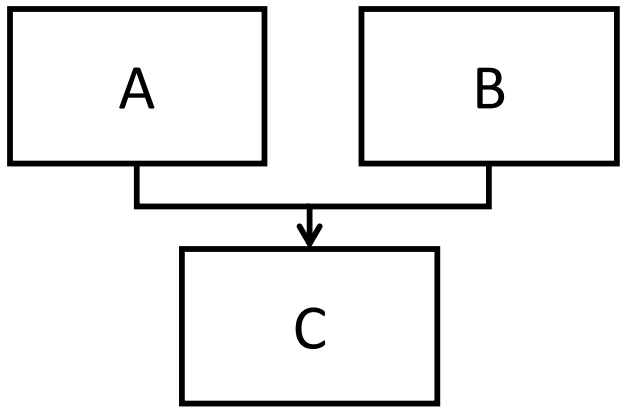
class DA extends HRA
{
float da=2000;
}
```

```
class Science extends DA
{
float bonous=2000;
public static void main(String args[])
{
Science obj=new Science();
obj.total_sal=obj.salary+obj.hra+obj.da+obj.bonous;
System.out.println("Total Salary is:"+obj.total_sal);
}
}
```

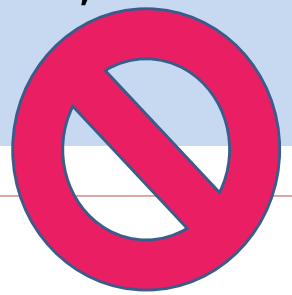
Output

```
Total Salary is: 37000.0
```

3. Multiple Inheritance



- If a class is derived from more than one class then it is called **multiple inheritance**.
- Here, class **C** is derived from two classes, class **A** and class **B**.
- It is not supported in Java

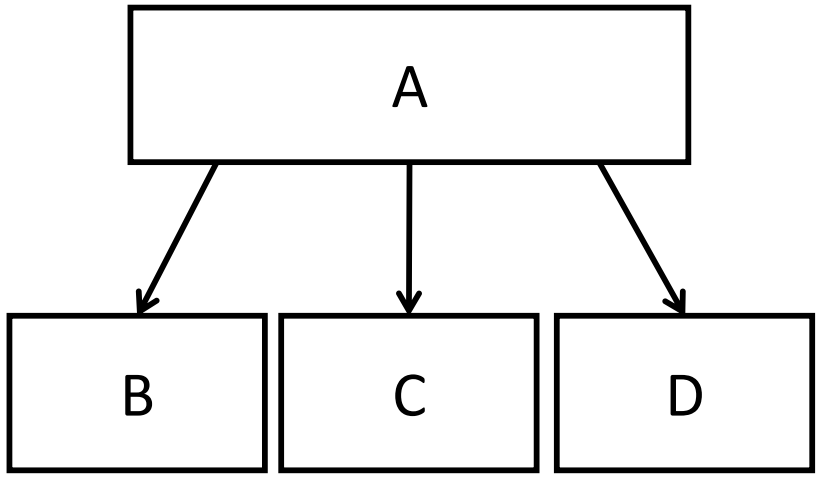


Example:

```
class Liquid
{
   //content of Liquid class
}
class Fuel
{
   //content of Fuel class
}
```

```
class Petrol extends Liquid,
Fuel
{
      //content of Petrol class
}
```

4. Hierarchical Inheritance



- If one or more classes are derived from one class then it is called **hierarchical inheritance**.
- Here, class **B**, class **C** and class **D** are derived from class **A**.

Example:

```
class Animal
{
//content of class Animal
}
class Elephant extends public Animal
{
//content of class Elephant
}
```

```
class Horse extends public Animal
{
//content of class Horse
};
class Cow  extends public Animal
{
//content of class Cow
};
```

4. Hierarchical Inheritance

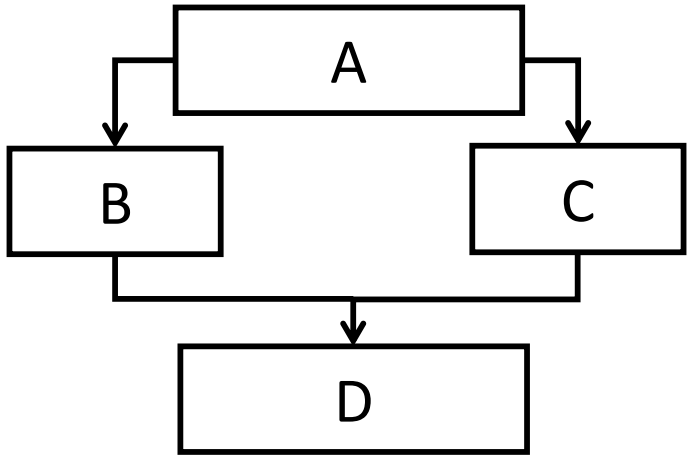
```
class Person {
public void show() {
System.out.println("I am a Person");
}
}
class Student extends Person {
public void show1() {
System.out.println("I am a Student who is Person ");
}
}
class Teacher extends Person {
 public void show2() {
System.out.println("I am a Teacher who is a Person");
}
}
```

```
class Doctor extends Person {
public void show3() {
System.out.println("I am a Doctor who is a Person");
}
}
public class HierarchicalInheritance {
public static void main(String args[]) {
Teacher teacher = new Teacher();
Student student = new Student();
Doctor doctor = new Doctor();
student.show();
student.show1();
teacher.show2();
doctor.show3();
}
}
```

Output:

```
I am a Person
I am a Student who is Person
I am a Teacher who is a Person
I am a Doctor who is a Person
```

5. Hybrid Inheritance



Hybrid Inheritance is also not supported in Java

- It is a combination of any other inheritance types. That is either multiple or multilevel or hierarchical or any other combination.
- Here, class **B** and class **C** are derived from class **A** and class **D** is derived from class **B** and class **C**.
- class **A**, class **B** and class **C** is example of **Hierarchical Inheritance** and class **B**, class **C** and class **D** is example of Multiple Inheritance so this hybrid inheritance is combination of Hierarchical and Multiple Inheritance.

Practice Program

- Create a class student that stores roll_no, name. Create a class test that stores marks obtained in five subjects. Class result derived from student and test contains the total marks and percentage obtained in test. Input and display information of a student.

